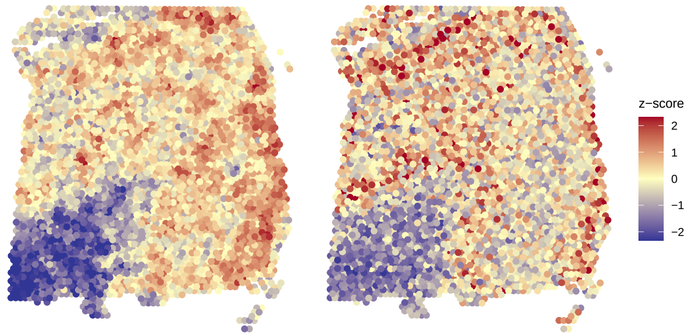
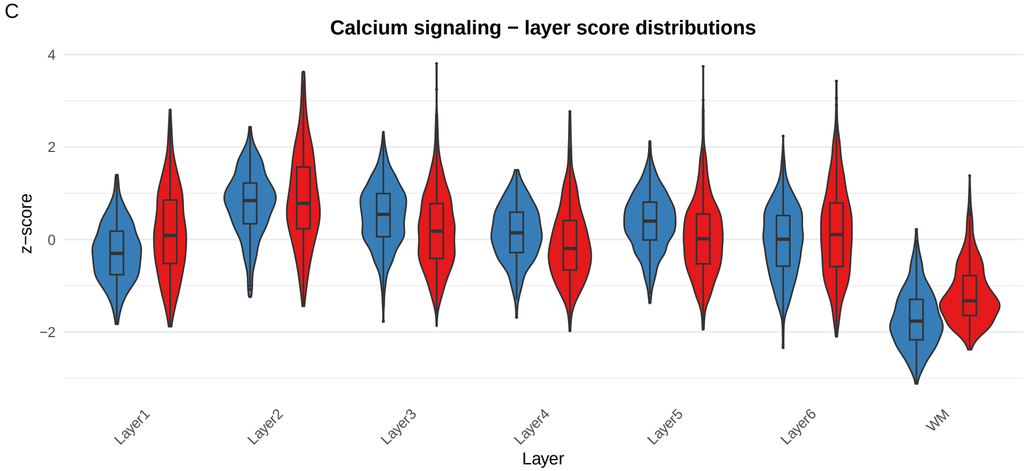
A


Calcium signaling − sGSVA

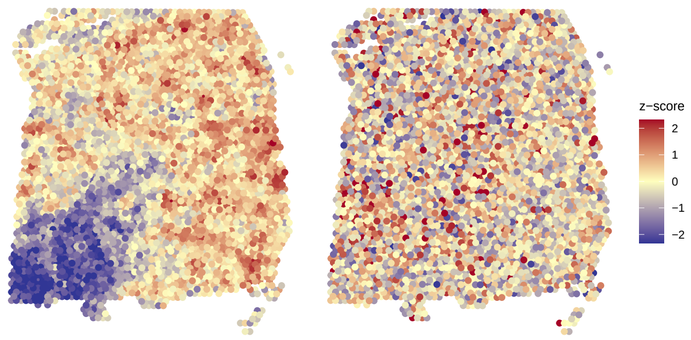
B

Calcium signaling − GSVA
z−score
2
1
0
−1
−2

C


Calcium signaling − layer score distributions
z−score
4
2
0
−2
Layer1
Layer2
Layer3
Layer4
Layer5
Layer6
WM
Layer

D


Axon guidance − sGSVA

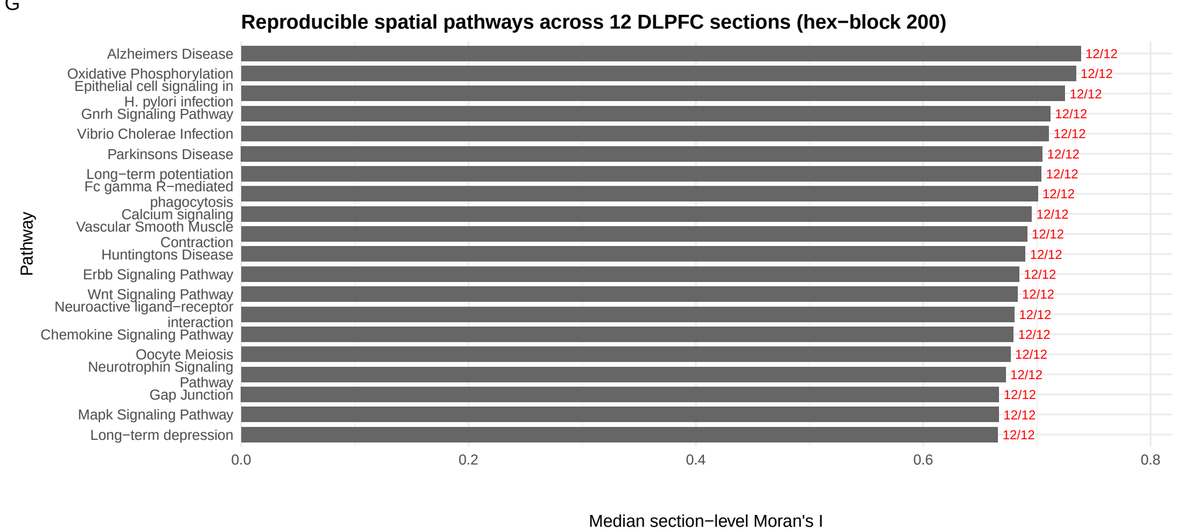
E

Axon guidance − GSVA
z−score
2
1
0
−1
−2

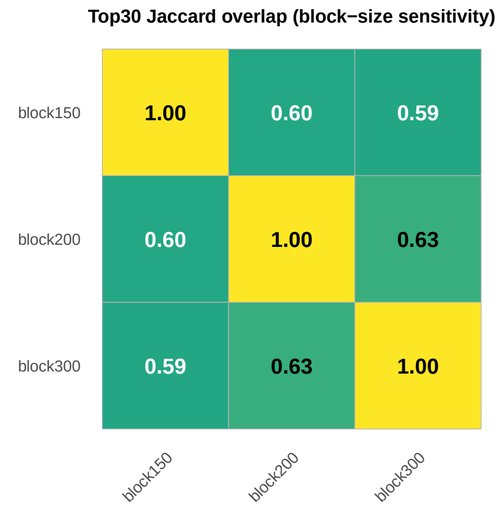
F


Axon guidance − layer score distributions
z−score
4
2
0
−2
Layer1
Layer2
Layer3
Layer4
Layer5
Layer6
WM
Layer
method
sGSVA
GSVA
Value
1.00
0.75
0.50
0.25
0.00

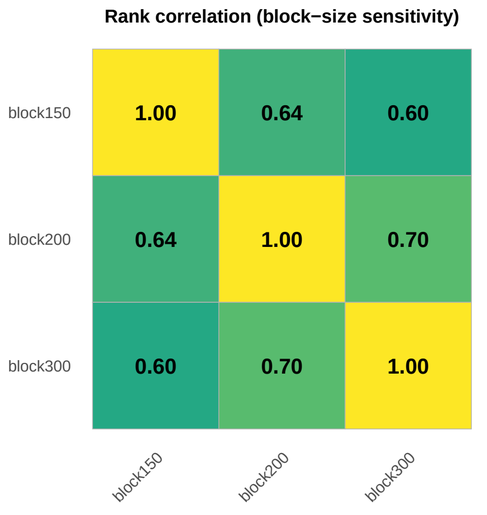
G


Reproducible spatial pathways across 12 DLPFC sections (hex−block 200)
Pathway
Alzheimers Disease
Oxidative Phosphorylation
Epithelial cell signaling in H. pylori infection
Gnrh Signaling Pathway
Vibrio Cholerae Infection
Parkinsons Disease
Long−term potentiation
Fc gamma R−mediated phagocytosis
Calcium signaling
Vascular Smooth Muscle Contraction
Huntingtons Disease
Erbb Signaling Pathway
Wnt Signaling Pathway
Neuroactive ligand−receptor interaction
Chemokine Signaling Pathway
Oocyte Meiosis
Neurotrophin Signaling Pathway
Gap Junction
Mapk Signaling Pathway
Long−term depression
12/12
0.0
0.2
0.4
0.6
0.8
Median section−level Moran's I

H


Top30 Jaccard overlap (block−size sensitivity)
block150
block200
block300
1.00
0.60
0.59
0.60
1.00
0.63
0.59
0.63
1.00

I


Rank correlation (block−size sensitivity)
block150
block200
block300
1.00
0.64
0.60
0.64
1.00
0.70
0.60
0.70
1.00